

PROBLEM SET 3

Due date: April 5, 2013 at the recitation session

1. Consider a homogenous good Bertrand duopoly (2 firms) with a market demand

$$Q_D = p^{-1/2}, \quad p \in [0, \infty)$$

Suppose that the firms can produce each unit at 0 marginal cost. The low price firm captures all the demand and the firms split demand equally if they charge the same price.

- (a) Does this game have pure strategy Nash equilibria? If yes, what are they?
 - (b) Consider now an equilibrium, in which the two firms adopt mixed strategies. Show that there exists a symmetric mixed-strategy Nash equilibrium of this game, in which each firm makes an expected profit of $\tilde{\pi}$. Hint: find the equilibrium CDF $F(p)$ and its support $[\underline{p}(\tilde{\pi}), \bar{p}(\tilde{\pi})]$ that generates this expected payoff.
2. Consider the game depicted in Figure 1, which is a three player game of perfect information in extensive form.

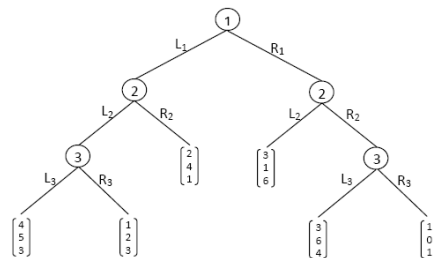


Figure 1

- (a) Find all of the subgame-perfect Nash equilibria in pure strategies.
- (b) Find all Nash equilibrium outcomes that do not entail mixed strategies (Note: there may be several strategy profiles which support a particular Nash equilibrium outcome, you only need to find one profile for each NE outcome)

3. Consider the game depicted in Figure 2.

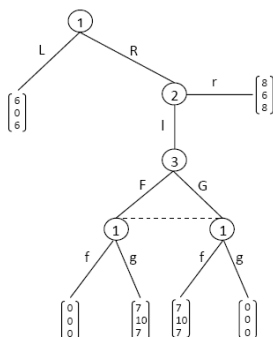


Figure 2

- Find all Nash equilibria between player 1 and 3 at the third stage. Calculate the corresponding payoffs.
 - Solve for the Subgame Perfect Nash Equilibrium of the three stage game. Show that in all SPNE player 1 plays R.
 - Subgame perfection supposes not only that the players expect Nash equilibria in all subgames, but also that all players expect the same equilibria. If this assumption is relaxed, it may be reasonable for player 1 to play L. What 'story' justifies this choice?
4. This is a two-person dynamic game of perfect information that is played as follows. There is a $K \times K$ grid of points X where $K \geq 2$. The points in the grid are numbered in matrix fashion, so that x_{11} is the upper left corner and x_{KK} is the lower right corner. Player 1 begins by taking the first chomp (bite) out of the grid. Specifically, he selects a point in the grid $x_{kl} \in X$, and removes it along with all the points above and to the left of it (i.e., all points $x_{ij} \in X$ with $i \leq k$ and $j \leq l$). Player two then moves, taking the next chomp out of the remaining points in the grid, and so on. The loser is the player who takes the last chomp (i.e., the one who chomps points x_{KK}).
- Prove that player I wins this game in every SPNE. (Hint: suppose by way of contradiction that II possesses a winning strategy and consider the possible subgames that start following II 's first chomp.)
 - Find a 'rule' player I can use that ensures he always wins Chomps. Why is this 'rule' not a strategy?
 - Draw the game tree corresponding to Chomp(2,2). (Assume that the payoff to winning is 1 and the payoff to losing is -1 . Assume also that if only x_{22} remains, then the game has reached a terminal node.)
 - Let $a_{it}(\cdot)$ denote the t -th action taken by player i (i.e. the node he chomps) as a function of the points that have been chomped at that stage in the game. Using this notation, specify the SPNE strategies in Chomp(2,2). For instance, it might be the case that $a_{11}(\emptyset) = x_{11}$.